Independent-Protocol Recovery of Resolved–Null Coupling in CTMT:

A Synthetic Coil, Navier, Seismic, and Degradation Battery

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Abstract

This note reports a synthetic independent-protocol recovery battery for the CTMT resolved–null covariance coupling C_{RN}. The purpose is not to establish physical universality. The purpose is to test whether a controlled coupling can be recovered across independent observation protocols: coil sensor-layout and bandwidth changes, Navier-like observable choices, seismic catalogue pipelines, and deliberate degradation by missing samples, temporal jitter, low-pass filtering, and covariance-estimator shrinkage. The results support a conservative conclusion: C_{RN} is recoverable when the observation protocol preserves enough of the resolution geometry, degrades continuously under filtering and estimator changes, and fails when the protocol changes the resolved sector too strongly.

1 Purpose

The local CTMT resolution geometry is

$$\mathcal{G} = (O; R, N, \Sigma), \qquad R = \operatorname{Im} J, \qquad N = \ker J^\top.$$

Relative to $O = R \oplus N$,

$$\Sigma = \begin{pmatrix} \Sigma_R & C_{RN} \\ C_{RN}^\top & \Sigma_N \end{pmatrix}.$$

The object tested here is not merely the magnitude $\|C_{RN}\|_F$, but the recovery of coupling orientation and block geometry across independent protocols.

2 Diagnostics

For a base protocol and a comparison protocol, the battery reports

$$d_R = \frac{\|P_{R,0} - P_{R,1}\|_F}{\|P_{R,0}\|_F + \|P_{R,1}\|_F},$$

$$d_C = \frac{\|C_{RN}^{(0)} - C_{RN}^{(1)}\|_F}{\|C_{RN}^{(0)}\|_F + \|C_{RN}^{(1)}\|_F},$$

and

$$\cos(C_{RN}^{(0)}, C_{RN}^{(1)}) = \frac{\langle C_{RN}^{(0)}, C_{RN}^{(1)} \rangle_F}{\|C_{RN}^{(0)}\|_F \|C_{RN}^{(1)}\|_F}.$$

A protocol is counted as recovered if $\cos(C) > 0.85$ and $d_C < 0.25$ without rank or sector failure. It is counted as degraded but recognizable if $\cos(C) > 0.65$ and $d_C < 0.45$.

3 Battery classes

(a) Same synthetic coil observed with different sensor layouts and bandwidths.

(b) Same Navier-like system observed through velocity, vorticity, pressure, and energy-density channels.

(c) Same synthetic seismic region observed through independent catalogue-like preprocessing pipelines.

(d) Deliberate degradation of the coil chart through missing samples, temporal jitter, low-pass filtering, and covariance shrinkage.

4 Results

Domain	Case	d_R	d_C	$\cos(C)$	Verdict
coil	coil_radial_sparse	0.061	0.225	0.899	COUPLING_RECOVERED
coil	coil_mixed_offaxis	0.057	0.230	0.895	COUPLING_RECOVERED
coil	coil_low_bandwidth	0.000	0.232	0.893	COUPLING_RECOVERED
navier	navier_vorticity	0.707	0.375	0.719	PROTOCOL_CHANGED_SECTOR_GEOMETRY
navier	navier_pressure	0.362	0.385	0.704	PROTOCOL_CHANGED_SECTOR_GEOMETRY
navier	navier_energy	0.648	0.386	0.703	PROTOCOL_CHANGED_SECTOR_GEOMETRY
seismic	seismic_catalog_B	0.055	0.409	0.666	COUPLING_DEGRADED_BUT_RECOGNIZABLE
seismic	seismic_catalog_C	0.048	0.381	0.711	COUPLING_DEGRADED_BUT_RECOGNIZABLE
degradation	missing_20pct	0.321	0.732	-0.072	COUPLING_NOT_RECOVERED_OR_ARTIFACT
degradation	temporal_jitter	0.000	0.607	0.268	COUPLING_NOT_RECOVERED_OR_ARTIFACT
degradation	low_pass_filter	0.000	0.657	0.193	COUPLING_NOT_RECOVERED_OR_ARTIFACT
degradation	shrinkage_estimator	0.000	0.111	0.998	COUPLING_RECOVERED

Table 1: Independent-protocol recovery battery. d_R reports sector drift, d_C reports resolved–null coupling distance, and $\cos(C)$ reports coupling-orientation recovery.

5 Interpretation

Proposition 1 (Synthetic independent-protocol observation)**.** *In the tested synthetic systems, resolved–null coupling was recovered when independent protocols preserved the Jacobian-induced resolution sectors sufficiently. Coupling was degraded but still recognizable under moderate filtering or estimator changes. Coupling recovery failed when the protocol substantially changed the resolved sector or observation map.*

The strongest conservative conclusion is that C_{RN} behaves as an observable-geometric invariant only relative to an admissible protocol family. The battery supports the protocol-invariance criterion but does not prove that nonzero coupling is physically universal.

6 Falsifiers

Falsifier 1 (Independent protocol failure)**.** *If independent admissible protocols for the same transport object produce incompatible C_{RN} orientation while R and N remain stable, then the coupling should be treated as estimator or protocol dependent.*

Falsifier 2 (Sector failure)**.** *If d_R or d_N is large, coupling comparison is not meaningful because the protocols no longer reconstruct the same resolution geometry.*

Falsifier 3 (Degradation discontinuity)**.** *If small loss, jitter, or filtering causes discontinuous coupling collapse, then the coupling is not robustly reconstructable.*

7 Non-claims

Non-claim 1. *This battery does not prove a physical interpretation of* C_{RN}.

Non-claim 2. *This battery does not show that all observation protocols should recover the same coupling.*

Non-claim 3. *This battery does not replace domain-specific covariance estimation or experimental validation.*

8 Conclusion

The battery sharpens the previous coupling claim. The relevant question is not whether $C_{RN} \neq 0$ in a single protocol, but whether the coupling is recovered across independent admissible protocols. In the synthetic tests, recovery occurs only when the observation protocol preserves sufficient resolution geometry. This supports the conservative CTMT reading: C_{RN} is a local resolution-geometric invariant under admissible protocols, while physical interpretation requires independent-protocol recovery.

References

[1] N. N. Čencov, *Statistical Decision Rules and Optimal Inference*, AMS, 1982.

[2] S. Amari and H. Nagaoka, *Methods of Information Geometry*, AMS/Oxford, 2000.

[3] C. R. Rao, *Linear Statistical Inference and Its Applications*, Wiley, 1973.

[4] R. A. Horn and C. R. Johnson, *Matrix Analysis*, Cambridge University Press, 2013.

[5] O. Ledoit and M. Wolf, A well-conditioned estimator for large-dimensional covariance matrices, *Journal of Multivariate Analysis*, 88(2):365–411, 2004.